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5.5 Similarity Transformations

Two figures are similar exactly when a sequence of **rigid motions** (translations, rotations, reflections — which preserve size and shape) and a single **dilation** (which scales size by k but preserves shape) maps one onto the other. Rigid motions preserve every length and angle; a dilation preserves every angle but multiplies every length by $|k|$. Because of this, a similarity transformation always preserves angle measure, and scales all lengths by the same overall factor — even a compound sequence of several dilations has a single **net scale factor** equal to the product of the individual factors. When $k = 1$, a similarity transformation is simply a **congruence transformation**. Reflections are the one type of rigid motion that reverses orientation (clockwise ↔ counterclockwise), while translations and rotations preserve it. This worksheet practises identifying, composing, and reversing similarity transformations on the coordinate plane.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Corresponding sides 8 and 2. Scale factor from larger to smaller?

2. Name the two types of transformations that make up a similarity mapping.

3. Under a similarity with $k = 3$, a side of length 5 becomes what length?

4. An angle measuring 110° maps under a similarity transformation to what measure?

5. A similarity transformation with $k = 1$ is also called what?

6. Perimeter 18 maps to perimeter 45. Find k.

7. Points $(2, 4)$ and $(4, 8)$ suggest what origin-centered scale factor?

8. Which type of rigid motion reverses orientation?

9. A side of 21 after a dilation with $k = \dfrac{1}{7}$ becomes what length?

10. A figure is dilated by $k = 0.5$, then rotated. Is the image similar to the original?

11. Corresponding sides 9 and 12. Find k from the first figure to the second.

12. Do similarity transformations always preserve angle measures?

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13. A figure is dilated about the origin by $k_1 = 1.5$, then dilated again about the origin by $k_2 = 4$. Find the net scale factor and the image of $(4, -2)$.

14. Point $(3, 7)$ is reflected over the x-axis, then dilated by $k = 3$ about the origin. Find the final image.

15. $\triangle DEF$ with $D(0, 0), E(3, 0), F(0, 5)$ maps to $D'(2, 4), E'(8, 4), F'(2, 14)$. Find the scale factor and describe a dilation-then-translation sequence that works.

✓ Answer Key

1. $\frac{1}{4}$
2. Rigid motions and a dilation
3. 15
4. $110°$
5. A congruence transformation
6. $\frac{5}{2}$
7. $k = 2$
8. Reflection
9. 3
10. Yes
11. $\frac{4}{3}$
12. Yes
13. Net $k = 6$; image of $(4, -2)$ is $(24, -12)$
14. $(9, -21)$
15. $k = 2$; dilate about the origin with $k = 2$, then translate by $(2, 4)$